

J. M. MURPHY.
SUPPORT FOR ELEVATOR CARS.
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1,392,962.

Patented Oct. 11, 1921.

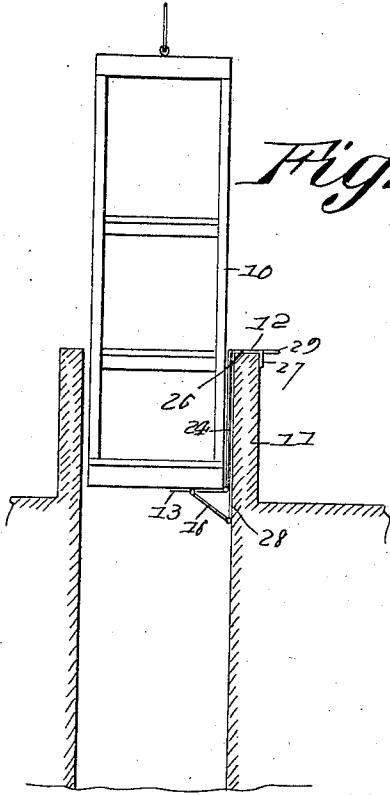


Fig. 1.

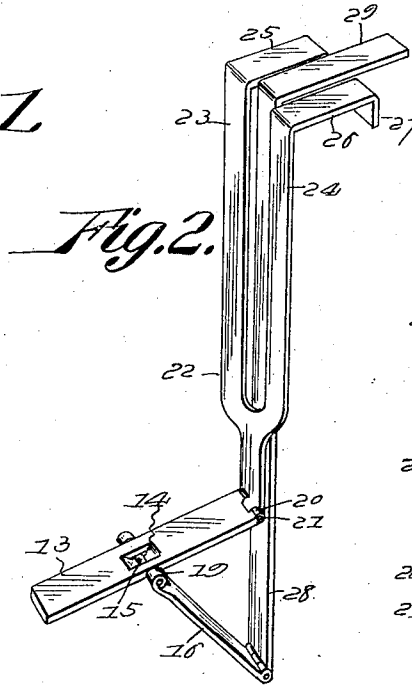


Fig. 2.

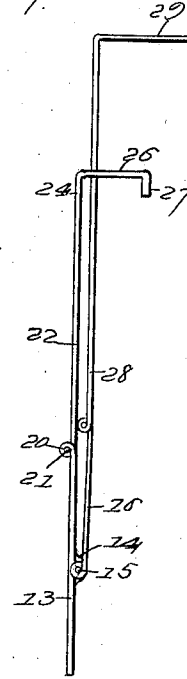


Fig. 3.

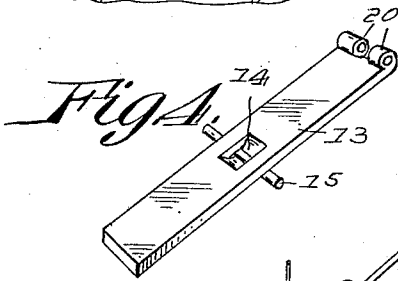


Fig. 4.

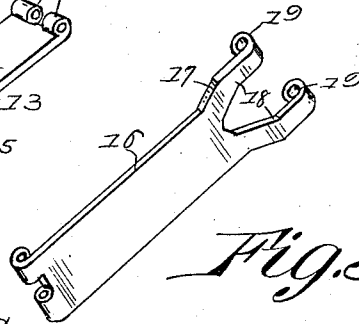


Fig. 5.

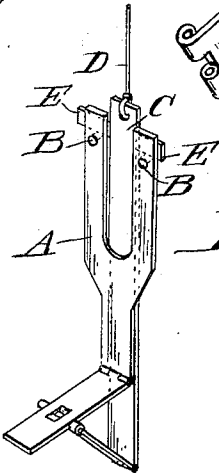


Fig. 6.

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SUPPORT FOR ELEVATOR-CARS.

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To all whom it may concern:

Be it known that I, JOHN M. MURPHY, a citizen of the United States of America, and resident of Shamokin, in the county of Northumberland and State of Pennsylvania, have invented certain new and useful Improvements in Supports for Elevator-Cars, of which the following is a specification.

This invention relates to supports for elevator cages or cars and has for its object the provision of novel means for holding such a cage in an elevated position; the said invention being more particularly intended for use in connection with battery chutes.

In the present equipment, the elevator when lifted out of the chute is placed on the ground and is liable to fall or be blown over, resulting in the loss of wet batteries, a condition which causes a short circuit in the track conductors.

The present invention provides for a device which may be supported over the edge of a chute, the elevator being then elevated and the device set to support the car, it being possible by this device to support the elevator at any of the shelves with relation to which the elevator operates.

A still further object of this invention is to produce an elevator car support which can be adjusted in a chute in a manner not to interfere with the movement of the car in the chute, whereas the said support can be manually operated and brought into operative position to support the car.

The invention is furthermore applicable for use in supporting dumb waiters and like hoisting mechanisms.

With the foregoing and other objects in view, the invention consists in the details of construction, and in the arrangement and combination of parts to be hereinafter more fully set forth and claimed.

In describing the invention in detail, reference will be had to the accompanying drawings forming part of this application wherein like characters denote corresponding parts in the several views, and in which—

Figure 1 illustrates a view in elevation of a car with the chute in section, showing the device embodying the invention applied thereto;

Fig. 2 illustrates a perspective view of the car supporting device;

Fig. 3 illustrates a side elevation thereof with the parts in different positions of adjustment;

Fig. 4 illustrates a perspective view of the supporting foot;

Fig. 5 illustrates a perspective view of the foot brace; and

Fig. 6 illustrates a perspective view of a modified construction.

In these drawings 10 denotes the elevator cage or car and this element will hereafter be referred to as a car which is intended to include any member adapted to travel in a chute for carrying commodities. The chute 11 is conventionally shown and it is to be understood that any casing or guide for the car will be comprehended within the said term, and 12 will be regarded as a sill at the chute opening through which opening the car is loaded or unloaded.

The car supporting device comprises a foot element 13 here shown as a plate, which, in the present embodiment of the invention has a cut-away and downwardly looped or bent portion 14, there being sufficient clearance between the downwardly looped portion and the under surface of the plate to receive a pintle 15 which pintle extends beyond the edges of the plate.

A prop or brace 16 has a bifurcated end 17, the furcations 18 of which are formed into eyes 19 in order that the brace or prop may be pivotally connected to the foot. This element 16 will be hereafter referred to as a brace.

The foot 13 has eyes 20 which receive a pintle 21 carried at the lower end of a hanger 22, the said hanger having parallel hanger elements 23 and 24 each of which has right angular extensions 25 and 26 respectively intended to overlies a sill 12 and by which the hanger is supported. Each right angular extension terminates in a flange 27 which extends down over the outer face of the sill to such a degree as to prevent dislodgment of the hanger from the sill. The brace 16 has its end remote from the pintle 15 pivotally connected to a bar 28 which is slidable between the extensions 25 and 26 of the hanger and it moves parallel with the side of the hanger when the parts are being adjusted. The bar 28 has an extension 29 standing at an angle to the bar which is likewise intended to overlies the sill between the extensions 25 and 26 and the relation of parts is such that the foot is supported in an approximately horizontal position under the car when the members 25, 26 and 29 are supported on the sill, but the

parts may be moved to the position shown in Fig. 3 by elevating the bar 28 to a position above the sill and when the parts are in this position, the foot, of course, is parallel with the wall of the chute and clear of the car, so that the car may travel in the chute without obstruction.

Assuming that the device is in place in a chute in the position in which it is shown in Fig. 3 and that a car has been elevated to the opening above the sill having the equipment, the operation of the bar by moving it downwardly will result in forcing the foot to the position shown in Fig. 2 and further movement of the bar downwardly will result in bringing it to the position shown in Fig. 1 where it would be effective to support the car, and of course a reverse movement of the bar 28 will result in returning the parts to the position shown in Fig. 3.

It has been found that a device made in accordance with this invention may be constructed of sheet metal almost wholly by the stamping process and that the cost of production is comparatively small and the device in use has proven strong and durable.

The modification shown in Fig. 6 is very similar to the construction of the device just described except that provision is made for its installation in new structures so that the device can be permanently attached in place. To that end, the member A, corresponding to the member 22 of the preferred form, is anchored permanently in place by fastenings B which are in the nature of pins or

bolts, and in this modified construction, the member C is slidable and may be provided with a flexible member D to facilitate its operation. The member C has a cross bar E which engages the pins or bolts B when in its lowermost position, and by this construction the necessity for having the right angular extensions 25, 26 and 29 of the other form is overcome.

I claim:

1. In a support for elevator cars, a hanger having means for its suspension, a foot pivotally connected to the hanger, a bar having movement parallel with the hanger and adapted to project therebelow.

2. In a support for elevator cars, a hanger having means for its suspension, a slidable bar, a foot pivotally connected to the hanger and a brace pivotally connected to the lower end of the bar and to the foot between its ends.

3. In a support for elevator cars, a hanger having means for its suspension, a bar slidably associated with the hanger and adapted to project therebelow, a foot pivotally connected to the hanger, and a brace pivotally connected to the bar and to the foot.

4. In a support for elevator cars, a hanger having angularly disposed extensions terminating in downwardly extending flanges whereby the hanger is suspended, a bar guided by the hanger for sliding movement therewith, a foot pivotally connected to the lower end of the hanger, and a brace pivotally connected to the bar and to the foot.

JOHN M. MURPHY.